

CHAPTER 11 : REFLECTION OF LIGHT

Topic:-

Make a kaleidoscope and make a presentation in the class about how it works.

Introduction:-

Light is a radiant form of electromagnetic energy that enables us to perceive the captivating colors and forms of the world around us. When light rays strike a smooth, polished plane mirror, they bounce back into the same medium obeying the fundamental **Laws of Reflection**: the angle of incidence equals the angle of reflection ($i = r$), and the incident ray, reflected ray, and normal all lie in the same geometric plane.

While a single plane mirror forms a single virtual image of an object, placing two or more plane mirrors inclined at an angle to each other produces a fascinating optical phenomenon called **Multiple Reflection**. The image formed by the first mirror acts as a virtual object for the second mirror, generating multiple symmetrical reflections.

In 1816, Scottish physicist Sir David Brewster utilized this principle to invent the **Kaleidoscope** (derived from Greek words meaning "observation of beautiful forms"). In this project, we constructed a functioning triangular kaleidoscope using three plane mirror strips, colorful broken glass bangles, and a cylindrical cardboard tube, and delivered an illustrated presentation in our science classroom explaining its optical mechanism.

❖ Core Optical Principles of the Kaleidoscope:-

- Three identical plane mirrors arranged at an equilateral angle of 60° to one another.
- Light undergoes repeated reflections between the inclined mirror surfaces.
- Produces breathtaking symmetrical hexagonal patterns that change constantly with rotation.

Aim & Kaleidoscope Setup:-

Aim: To fabricate a functional three-mirror kaleidoscope, demonstrate multiple reflection of light, and deliver an interactive classroom presentation explaining its optical geometry.

❖ Objectives of the Project:-

- To assemble three identical rectangular plane mirror strips inclined at 60° inside a protective tube.
- To build a rotating object cell containing colorful translucent glass pieces and beads.
- To derive and verify the mathematical formula for multiple images ($n = (360^\circ / \theta) - 1$).
- To explain how symmetrical hexagonal symmetry patterns are formed during class presentation.

❖ Materials Required for Construction:-

- Three rectangular plane mirror strips of identical size (length 15 cm, width 3.5 cm).
- Sturdy cylindrical cardboard mailing tube (length 18 cm, diameter ~ 5 cm).
- Clear transparent circular plastic disc, translucent butter tracing paper, and black chart paper.
- Assorted pieces of colorful broken glass bangles, beads, adhesive cellophane tape, and scissors.



Step-by-Step Construction Procedure:-

The construction of our demonstration kaleidoscope was accomplished through five systematic steps:

❖ *Fabrication Procedure:-*

- **Step 1 (Prism Formation):** Took three clean rectangular mirror strips. Placed them side-by-side with reflecting surfaces facing inward. Taped their outer back seams together firmly to construct a rigid hollow triangular prism with an equilateral cross-section (each angle exactly 60°).
- **Step 2 (Enclosing in Tube):** Wrapped the triangular mirror prism in bubble wrap or soft corrugated paper and slid it snugly into the cylindrical cardboard tube to prevent shifting.
- **Step 3 (Object Cell Construction):** At one end of the tube, fitted a clear circular transparent plastic disc touching the mirror edges. Placed several colorful broken glass bangle pieces and translucent beads onto this disc.
- **Step 4 (Diffusing Screen):** Closed the object chamber with a disc of translucent ground glass or tracing paper, leaving enough space for the glass bits to tumble freely when rotated.
- **Step 5 (Eyehole Cap):** Sealed the viewing end with a circular black cardboard disc having a central viewing peephole of 8 mm diameter. Decorated the outer cylinder with colorful paper.

Observations & Mirror Angle Formula Table:-

To demonstrate multiple reflection mathematically during our presentation, we tested various angles of mirror inclination using two hinged plane mirrors before presenting the 3-mirror kaleidoscope:

Angle Between Mirrors (θ)	Formula: $n = (360^\circ / \theta) - 1$	Calculated Images (n)	Observed Visual Symmetry
120°	$(360^\circ / 120^\circ) - 1 = 3 - 1$	2 Images	Simple dual reflection
90°	$(360^\circ / 90^\circ) - 1 = 4 - 1$	3 Images	Square four-fold symmetry
72°	$(360^\circ / 72^\circ) - 1 = 5 - 1$	4 Images	Pentagonal symmetry
60° (Kaleidoscope)	$(360^\circ / 60^\circ) - 1 = 6 - 1$	5 Images + 1 Object	Hexagonal six-fold symmetry
45°	$(360^\circ / 45^\circ) - 1 = 8 - 1$	7 Images	Octagonal radial symmetry
0° (Parallel Mirrors)	$(360^\circ / 0^\circ) - 1$	Infinite (∞) Images	Endless tunnel effect

At an angle of 60°, the original object together with its 5 reflected images forms a stunning, perfectly closed 6-fold radial pattern.

Scientific Analysis & Presentation Script:-

Below is an excerpt of our classroom presentation delivered in front of our science teacher and peers:

❖ Presentation Explanation Delivered to Classmates:-

"Dear classmates, look through the viewing peephole of this kaleidoscope while directing the other end toward a daylight window. When you gently rotate the cylinder, you see brilliant, endlessly shifting colorful geometric mandalas! But why do these patterns appear?"

*"The scientific secret lies in **Multiple Successive Reflections**. Inside the cylinder, three plane mirrors form an equilateral triangle where each angle is exactly 60° . When light enters through the translucent screen, it illuminates the tumbling glass pieces. Each mirror reflects the image of the object. But fascinatingly, the image reflected in Mirror 1 is immediately reflected again by Mirror 2, and then by Mirror 3!"*

"Because the angle is 60° , the formula $n = (360^\circ / 60^\circ) - 1$ produces 5 virtual images. Adding the original object, we obtain 6 symmetrical triangular wedges that seamlessly assemble into a complete 360° hexagon. As the glass pieces tumble randomly, the geometric configuration changes instantly, creating a brand new, never-repeating pattern!"

Everyday Applications & Discussion:-

The principles of multiple reflection and kaleidoscope patterns have exciting practical applications across diverse fields:

❖ Practical Industrial & Artistic Applications:-

- **Textile & Fabric Design:** Fashion designers and textile artisans use kaleidoscope patterns as infinite inspiration for creating unique motifs on sarees, carpets, and wallpapers.
- **Architectural Tiles & Glass Art:** Ceramic tile manufacturers utilize radial mirror symmetry to produce mesmerizing geometric floor and stained-glass window patterns.
- **Optical Instruments:** Multiple reflection principles form the foundation of periscopes in submarines, laser cavity resonators, and fiber optic communication.

❖ Peer Discussion & Clarifications Addressed:-

- **Peer Question:** "Why use colored glass bangles instead of opaque plastic beads?"

Answer: Translucent colored glass allows external light to pass through, creating luminous, glowing jewel-like colors through both light transmission and specular reflection.

What We Learned, Conclusion & References:-

❖ What We Learned From This Project:-

- Mastered the laws of reflection of light and the mathematical formula governing multiple reflections.
- Constructed a precision optical device using readily accessible, low-cost domestic materials.
- Developed engaging science presentation skills, demonstrating abstract physics concepts through tactile devices.

❖ Project Conclusion:-

Our kaleidoscope project provided an engaging practical demonstration of multiple reflections between inclined plane mirrors. By combining three mirrors at 60° , we created an instrument that converts chaotic glass fragments into orderly, symmetrical hexagonal art. The successful classroom presentation demonstrated that optics is not merely textbook ray diagrams, but a vibrant science that creates beauty all around us.

❖ Bibliography & References:-

- Maharashtra State Bureau of Textbook Production (Balbharati) – Standard 9 Science & Technology (Chapter 11: Reflection of Light).
- Sir David Brewster – *The Kaleidoscope: Its History, Theory, and Construction* (Classic Treatise).
- NCERT Class 9 & 10 Science – Light: Reflection and Refraction Modules.